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Inventory Optimization

Integrated Business Planning Based on I_SAPIBP2

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Inventory Optimization

SAP Integrated Business Planning for inventory recommends total inventory targets to maximize profit while buffering for uncertainty and maintaining customer service levels.

Overview

Note

For additional guidance and best practices for implementing inventory optimization, you can explore the latest version of content maintained at [SAP Best Practices for SAP Integrated Business Planning](#) .

Inventory optimization positions stock to absorb uncertainties, such as the following:

- Forecast error
- Demand variability
- Supply uncertainty

By using a set of stochastic mathematical algorithms, inventory optimization determines the lowest stocking cost possible for locations across an entire supply chain and provide stocking targets directly to planners at an item-location-time period level of granularity.

How Uncertainties Lead to Ineffective Stocking Levels

Uncertainties multiply across the supply chain due to the interdependency between suppliers, plants, warehouses, and customer sites. If you only consider one portion of a supply chain, you may hold too much safety stock at one location, which leads to increased holding costs, or hold too

little safety stock at one location, which leads to stock-outs that either cause lower customer service levels or increased cost due to expedited orders.

The following table lists some common variables which cause uncertainty that lead to ineffective stocking levels:

Common variables

Customer Side	Supply Side
Simultaneous internal and external demand varying for stock	Batch size requirements
Errors in demand forecasting	Late shipments
Demand that varies by season or time	Frozen planning windows
Over-forecasting or under-forecasting demand	Multiple supply sources
Outliers and unplanned spikes in demand	Seasonal supply sources
Variability in deliveries to warehouses	Regulatory QA production holds
Late changes in promotional event plans, such delayed execution or changes in the planned scope	Forecast error of raw material sourcing

Why Optimize Safety Stock?

Inventory optimization helps manage uncertainty in the supply chain by determining the amount of safety stock to hold at stocking locations within the supply chain. Safety stock absorbs uncertainty, allowing you to maintain customer service levels. Inventory optimization determines the optimal safety stock for each stocking location and recommends targets, based on the following:

- Demand from customers and demand uncertainty (direct demand) and the impact that has on internal replenishment orders (indirect demand and indirect demand uncertainty)
- Forecast error and its impact across the supply chain
- Lead times and their variability, replenishment frequency, and lot sizes
- Target service levels at customers

The resulting recommended inventory levels balance risk and cost appropriately. Not every location will see a reduction in inventory, but the entire supply chain will achieve a cost reduction of inventory while maintaining service levels.

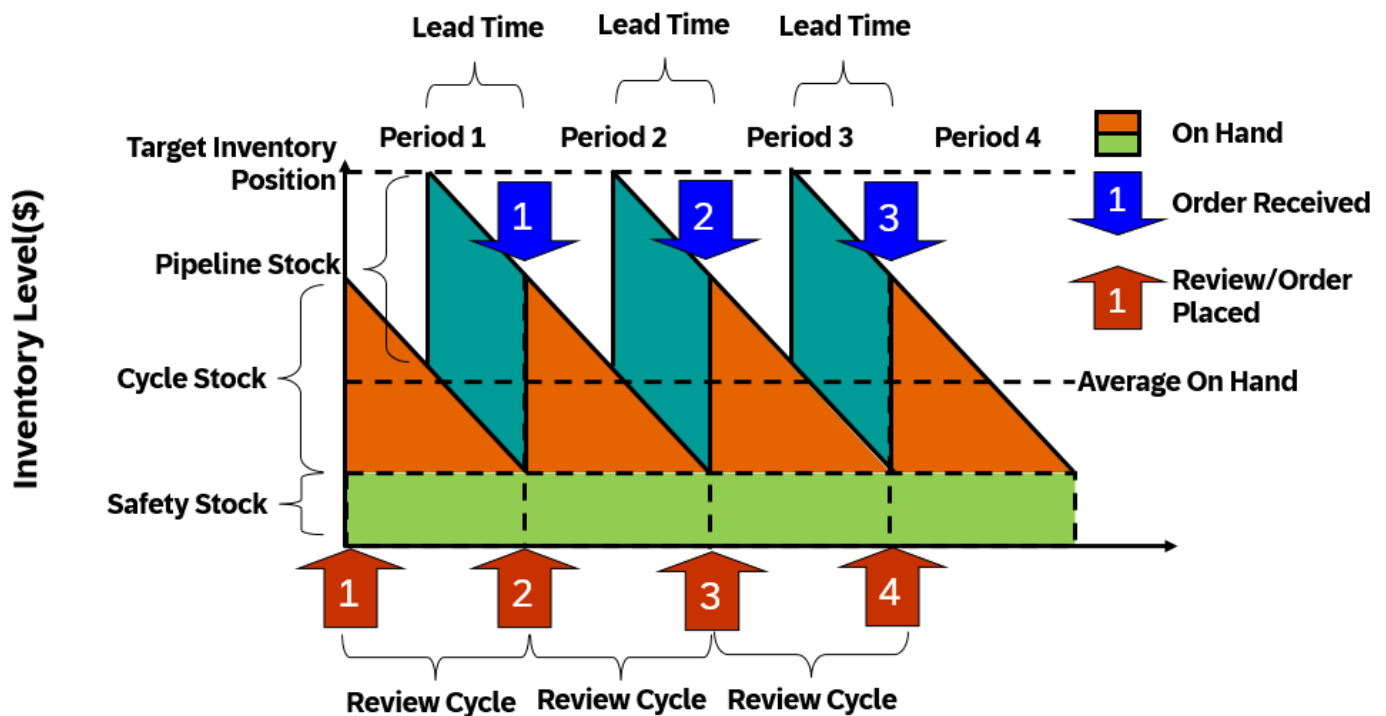
Optimal Inventory Placement

Optimal inventory placement is based on inventory forms and purposes, that is, on inventory components. Inventory components comprise the total inventory for a given product.

Calculating inventory components and their targets strengthens inventory optimization. Inventory components include estimates, targets, average quantities, and currency values. An example of inventory components include the following:

- On-hand stock – inventory that is physically in the supply chain, that is cycle stock and safety stock
- Cycle stock – inventory that is used to fulfill orders
- Pipeline stock – stock on order
- Safety stock – stock kept in reserve to meet unexpected demand and prevent stock-outs

The following figure illustrates the inventory replenishment process and the components involved:



Inventory Replenishment and Components

In the basic inventory ordering process, there is a periodic review where a planner looks at the current stock and what will be arriving, and compares that to forecast demand, which determines the committed inventory (on hand plus on order). If the committed inventory falls below the calculated target inventory position, you order up to the target inventory level.

Target Inventory Position

The correct target inventory position for a location is the amount of inventory needed to:

- Meet current demand

- Meet future demand until the next order is received
- Cover demand uncertainty

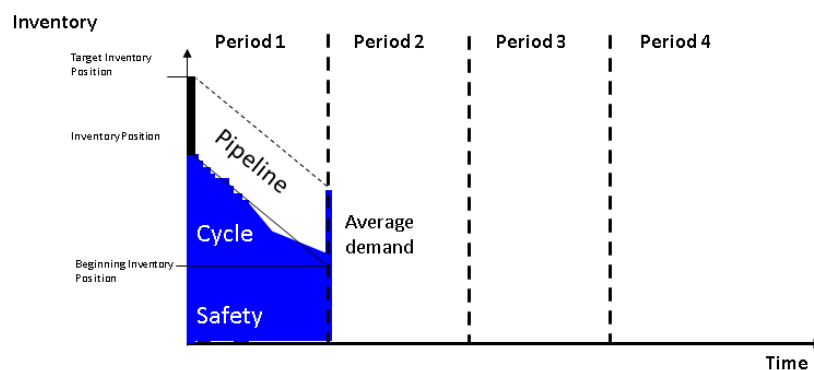
TIP is calculated based on forecasts and includes the following types of inventory:

- Pipeline stock
- Cycle stock
- Safety stock

Note

Target inventory position is independent of the committed inventory. For example, in a demand ramp-down situation, there may be enough inventory to cover future demand without placing an order.

The following figure illustrates the components of the target inventory position:

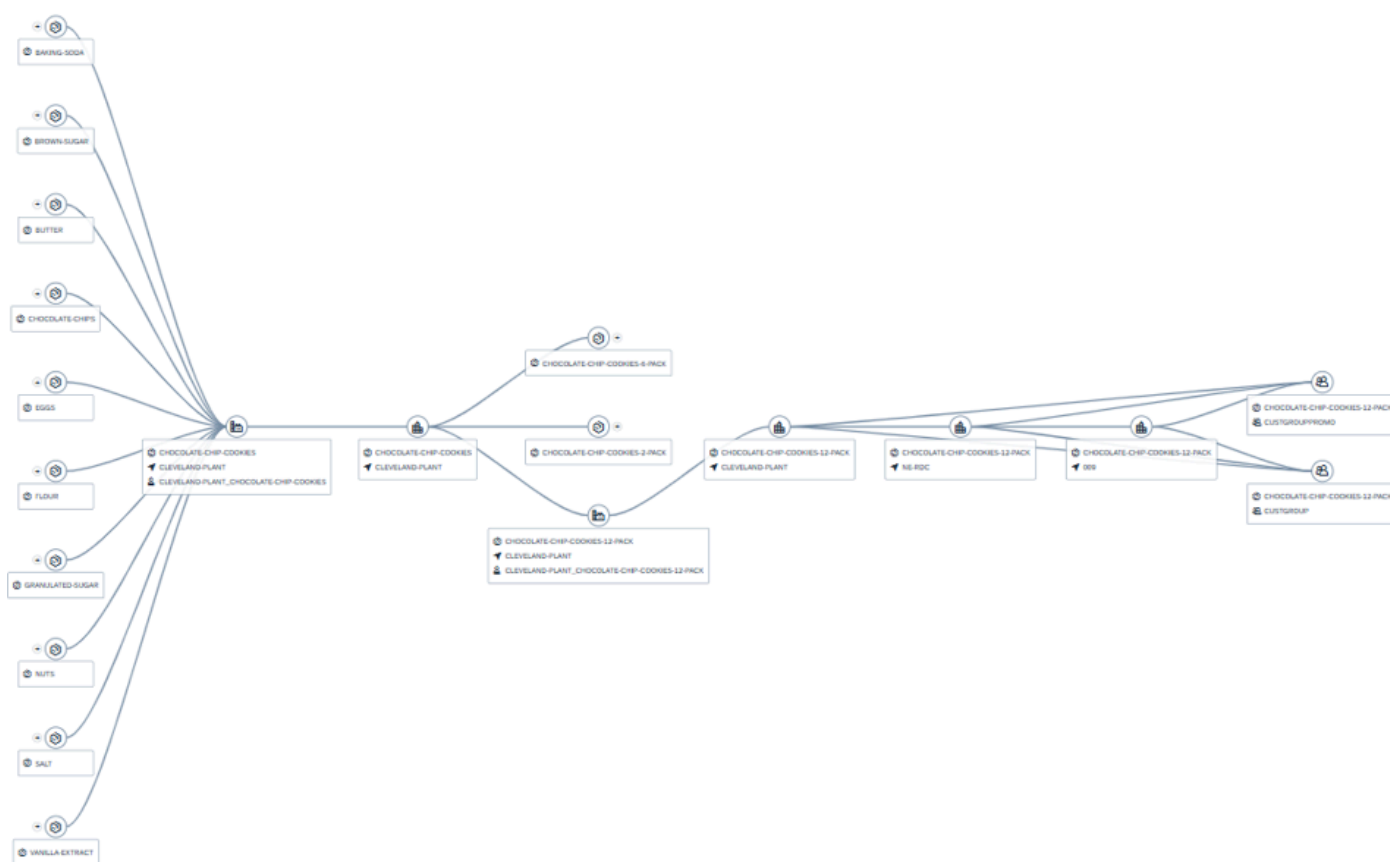


Target Inventory Position

Modeling a Supply Chain

A supply chain network modeled in SAP Integrated Business Planning for inventory drives optimization process and results. You can view the supply chain model using the **Supply Chain Network** app. For more information, see [Supply Chain Network](#).

The **Supply Chain Network** app renders a visualization of the complete supply chain, representing items, locations, and paths. The following figure illustrates a supply chain that has been visualized with the **Supply Chain Network** app:



Understanding the concept of how a supply chain is modeled helps with the understanding of how inventory optimization uses inputs from all the elements of a supply chain to determine inventory recommendations for the entire supply chain. The model is built automatically by the inventory operators in real-time, using source data from the ERP and source systems; it is not manually built by any user. The resulting supply chain model is a graphical representation of a supply chain network and the entities it contains. These entities, or nodes, represent supplier locations, customer locations, stocking, and non-stocking locations. Nodes are linked by sources of supply that represent the flow of inventory; either raw materials or finished goods.

In a typical supply chain model, a supplier node is where materials originate and a customer node is where finished products reach customers. In between supplier and customers, there are a number of other nodes that represent other locations where inventory resides as it flows from supplier to customer, such as manufacturing plants, warehouses, and docks. The following table describes these nodes:

Node Type	Description
Stocking node	A location where inventory is carried and that has a demand stream. Stocking nodes can be the following: Internal stocking nodes have internal demand streams, for example, a

Node Type	Description
	warehouse supplying raw materials to a manufacturing plant. • Customer-facing stocking nodes have customer-facing demand streams, for example, a location that supplies a store. • Hybrid nodes have both internal and customer-facing demand streams, for example, a location that ships finished goods to customer nodes, but also ships to an internal node to create multipacks.
Non-stocking node	Locations that do not hold any inventory, for example, production plants or cross-docks. Non-stocking nodes are flow-through points.
Process point	Not physical locations, but logical points along the supply chain model where processing of inventory occurs, for example, the creation of bills of material (BOMs).

Once you provide the input data, you can run inventory optimization to create the network topology and determine the optimal amount of safety stock to carry at stocking locations across the supply chain.

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